A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 40

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark for the point below. **1** Direct measurements of past conditions cannot be taken, so indirect evidence that is related to the condition must be used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
01.2	**1 mark each** 1 mark for each of the 4 points below. **1** Age / date of the sample (radioisotope dating) **1** Past temperatures **1** Composition of the past atmosphere (eg CO_2 / methane concentrations) **1 Temperature:** more magnesium is incorporated into calcite at higher temperatures **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Movement of continents changed the position of land masses / latitude and therefore the climate and habitats present **1** Ocean currents redistribute heat and nutrients – changes in currents (eg as continents moved) altered climates, so past conditions cannot be explained without knowing how the systems worked **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** There was no oxygen / ozone layer so high levels of UV radiation reached the land surface – water absorbed the UV and protected organisms in the oceans **1** Warm temperatures around the vents and a rich mix of chemicals / increasingly complex organic molecules in the sea made biological reactions more likely **1** The atmosphere contained toxic gases such as ammonia – the early Archaea were anaerobic so did not need oxygen **1** Water acted as the solvent for reactions / transport **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • Points after the first 3 - the question asks for 3, so a list of extras cannot pick up the marks. • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-9; AQA spec 3.1.1.1-3.1.1.3.	
02.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 600 + (90 + 150) - (220 + 80) = 540 **1** Stock fell by 60 so the catch exceeds the MSY / unsustainable **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (thousand tonnes) is already printed on the answer line. • **MSY** → the largest catch that can be taken without reducing the stock **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks. • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
02.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** A + G - M = 90 + 150 - 80 = 160 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (thousand tonnes) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
02.3	**1 mark each** 1 mark for each of the 3 points below. **1** Catch is landed and weighed at ports so it is recorded **1** Other terms need data on fish in the sea: populations are mobile (crossing national boundaries), very unevenly distributed **1** Representative sampling of a large enough proportion of the area is impractical / expensive – survival of eggs and larvae varies greatly between years **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.4	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 CPUE:** if more effort (boat-days, engine power, net size) is needed to land the same catch, the population is declining even though total catch is maintained – a better measure than catch size which technology can prop up **2** Falling mean size / age (from scale or otolith growth rings) shows older adults are being removed faster than replaced and the catch depends on small young fish **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 349-351; AQA spec 3.5.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 4 points below. **1 Energy:** low energy-density renewable (solar) / non-renewable fossil fuels **1 Materials:** cyclic - wastes of one process are raw materials of the next / linear - used once then discarded, depletion **1 Wastes:** biodegradable, non-toxic or do not build up / toxic, persistent (eg PCBs, organochlorines, toxic metals) that bioaccumulate **1 Temperature:** low / high (Haber process, incineration) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biodegradable** → broken down by decomposers • **persistent** → does not break down / stays in the environment for a long time • **incineration** → burning waste, usually with energy recovery **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	4
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Abundant simple raw materials build into monomers and then a wide range of polymers (carbohydrates, proteins) **1** They are re-used in cycles so there is no depletion of raw materials or build-up of wastes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.3	**2 marks** 2 marks for the point below; award fewer where it is only partly made. **2** Use often disperses the minerals (eg in products spread over the world / corrosion) **OR** produces mixtures of materials that are difficult to separate, so sustainable exploitation is reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2

Do not write outside the box

Question	Marking guidance	Marks
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** They cannot easily be removed from end-of-life appliances and are not biodegradable **1** Natural wastes are non-toxic / broken down to raw materials for other processes, whereas persistent toxins may bioaccumulate and biomagnify **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers • **persistent** → does not break down / stays in the environment for a long time	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.2-3.6.3.	
04.1	**1 mark each** 1 mark for each of the 2 points below. **1** Overall decline from 62% to 22% **1** Sharp falls in 1998 (to 35%) and 2016 (to 18%) coinciding with bleaching events, with partial recovery after each **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.2	**1 mark each** 1 mark for each of the 4 points below. **1** The polyps expel their symbiotic zooxanthellae so the coral loses colour – bleached corals are alive but grow very slowly and may die if algae do not return **1** Raised water temperature (climate change / El Niño) **1 Pollution:** sewage, pesticides, cyanide, sunscreen – low oxygen from zooplankton blooms after overfishing **1** High light / solar activity – ocean acidification hinders skeleton formation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **symbiotic** → a close relationship in which both species benefit **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4

Do not write outside the box

Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 3 points below. **1** Coastal construction and dredging increase sedimentation / turbidity covering and killing polyps **1** Sewage and fertiliser nutrients cause algal growth that shades corals and prevents egg release **1** Physical damage from anchors, divers, souvenir collection – loss of mangroves / seagrass nurseries that trap sediment **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	3
04.4	1 mark for the point below. **1** Protected breeding populations grow larger and produce surplus young / adults that spill over into fished areas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 60-63; AQA spec 3.1.2.3.4.	
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** The scatter / variability of results around each mean – about 68% of values lie within one SD of the mean **1** The SDs are large relative to the difference in means (1.9 g) so the ranges overlap greatly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean	2
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** t-test **1** Comparing two means of measured (continuous) data **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **t-test** → a test for a difference between the means of two normally distributed sets **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	2

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** There is a 30% probability that the difference occurred by random chance **1** Above the 0.05 (5%) level so the difference is not significant – the null hypothesis (no difference) is accepted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **random** → positions chosen using random numbers so the choice is not biased • **null hypothesis** → a statement that there is no significant difference or correlation **Do not credit** • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	2
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Two correlated variables may both be caused by a third variable that was not investigated **1** Significance only shows the result is unlikely to be chance – a mechanism / controlled experiment is needed for causation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **statistically significant** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **correlation** → a relationship in which the two variables change together • **causal** → one variable actually causing the change in the other **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test	2
05.5	**1 mark each** 1 mark for each of the 2 points below. **1** Increase the number of replicates (reduces the effect of variability) **1** Standardise other variables (soil, water, light) more closely to reduce scatter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Environmental DNA (eDNA) analysis	1

Do not write outside the box

Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Transmitters are too heavy for small birds / their batteries would not last for a long-term study **1** Geolocator tags are small and only record time and light levels – latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Each separate (regional) population has its own unique gene pool **1** DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	
07.1	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 Rotation:** each crop has its own pests – growing a different crop each year (4-5 year cycle) means pests left at the end of a season die before that crop returns, so populations cannot build up early **up to 2** Beetle banks / hedgerows provide habitat and overwintering sites for natural predators (ground beetles, ladybirds) that eat aphids **up to 2** Rotation also evens out nutrient demands and includes legumes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
07.2	**1 mark each** 1 mark for each of the 3 points below. **1** Integrated control gives almost the same yield as chemical control (18.1 vs 18.5) with far fewer applications (1-2 vs 6) **1** It retains most of the biodiversity of cultural control (26 vs 31 species) whereas chemical control alone leaves only 8 **1** Cultural control alone loses about 12% yield – integrated control combines cultural methods, resistant varieties and carefully timed specific non-persistent pesticides only when essential - best balance of yield, cost and impact **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3

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Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 3 points below. **1** Large numbers of males are sterilised with gamma radiation and released **1** Females of many insects mate only once and store sperm, so mating with a sterile male produces no offspring – enough sterile males reduce fertile matings below replacement level (screw worm fly, Mediterranean fruit fly) **1** Only the target species' females mate with them so non-target species are unaffected – fails if sterilised males behave abnormally or re-colonisation occurs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 314-318; AQA spec 3.5.1.1.	
08.1	**1 mark each** 1 mark for each of the 3 points below. **1 Diversity:** HEP (wet winters), solar (summer) and wind peak at different times so supply is more reliable - diverse systems resist change **1** No single source could meet demand all year – battery storage on the island grid smooths fluctuations **1** Calm, dry, dull periods still leave gaps so diesel generators provide security (fossil fuel use minimised, not yet eliminated) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3

Do not write outside the box

Question	Marking guidance	Marks
08.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Community ownership (Isle of Eigg Trust with Scottish Wildlife Trust) managing the island for people and wildlife **1** Wood biomass heating – insulation such as straw-bale housing – some electric vehicles **1 Habitat protection and creation:** fencing rare plants from grazing, planting indigenous woodland (rowan, birch, willow) – eagles and hen harriers nesting **1** Recyclables shipped off – garden / food waste composted – low-impact food production – beach litter collection **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainability** → meeting present needs without stopping future generations meeting theirs **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Small groups can organise, agree and act quickly – decisions benefit people who see the results directly **1** Resources and wastes are local and visible so feedback is immediate, unlike cities supplied from far away **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** Renewable energy is abundant but geographically and temporally variable – a continental grid moves electricity from areas of surplus (desert solar, offshore wind) to areas of demand **1** DC grids do not need every generator synchronised in phase as AC grids do, so coordination across many countries is easier **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 384-389; AQA spec 3.6.4.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 2 points below. **1** Yield increases as inputs increase **1** But each extra unit of input produces a smaller increase in yield than the previous one (the curve flattens / law of diminishing returns) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1 Second unit:** 70 - 50 = 20 units **1 Fifth unit:** 94 - 90 = 4 units **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (units) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** The first units of input on each area give the largest yield increases **1** Concentrating inputs means later units add very little, whereas the same inputs spread thinly all work at the steep part of the curve **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** Some areas have much greater potential productivity so investment there is sensible **1** Affluent societies can afford to buy more inputs while poorer communities that would benefit most cannot afford them (economic reasons) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.5	**1 mark each** 1 mark for each of the 2 points below. **1** Yield per unit area is very high because of large inputs **1** But the extra yield per unit of input is small, so efficiency (yield per unit input / energy ratio) is low **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-325; AQA spec 3.5.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Resources:** wood, fibres (cotton, wool, silk), oils, fuels (wood, biofuels), food – new food species (morama bean, perennial cereals, eland, bison) better adapted to local conditions • **Biomimetics:** vehicle design, architecture, structures, adhesion, materials, ultrasound • Medicines from plants and animals (examples) – most species unstudied - argument for habitat conservation • Physiological research using wildlife species – wildlife as pest control agents (ladybirds, Encarsia, Cactoblastis, Bt) • **Genetic resources:** crop wild relatives, Vavilov centres, gene pools • **Ecosystem services:** atmospheric composition, hydrological and biogeochemical cycles, soil maintenance, pollination, seed dispersal, habitat provision • Ethical / aesthetic / recreational arguments – the unknown value of species not yet investigated (precautionary principle) • **Evaluation:** which arguments are most persuasive to decision makers – economic valuation vs intrinsic value – judgement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **biodiversity** → the variety of life: habitat, species and genetic diversity **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	9
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Human society depends on other species first as resources: wood, fibres such as cotton, wool and silk, oils, fuels and, above all, food, with wild species such as the morama bean, perennial cereals, eland and bison offering new crops and livestock better adapted to local conditions. Biomimetics copies adaptations into vehicle design, architecture, structures, adhesives, materials and ultrasound; medicines such as taxol, quinine and AZT came from plants and animals, and physiological research on marsupials, hippos and sponges continues to yield medical insights. Wild species are pest-control agents - ladybirds, *Encarsia*, *Cactoblastis*, *Bacillus thuringiensis* - and crop wild relatives in Vavilov centres hold the genes future crops will need.	
	Less visibly, other species provide the services that keep the planet habitable: regulating atmospheric composition, driving the hydrological and biogeochemical cycles, maintaining soil, pollinating crops, dispersing seeds and providing habitat. Most species have never been investigated, so their value is unknown, and there are ethical, aesthetic and recreational reasons for conservation that owe nothing to usefulness.	
	Which arguments persuade decision-makers? Economic valuation of resources and services speaks the language of policy and has been most effective, but it undervalues the services that go unnoticed because they operate in dynamic equilibrium and it can never capture the value of species not yet studied. The strongest case combines the precautionary logic of unknown value with the intrinsic argument: conserving biodiversity protects both what we know we need and what we cannot yet know.	
	Weaker answer - Level 1 (3 marks)	
	Biodiversity is important for humans in many ways. We get food, wood and cotton from plants and animals. Many medicines come from plants. Bees pollinate our crops so we need to conserve them. Animals and plants are also nice to look at and people like to visit nature. We should conserve biodiversity because we might need species in the future.	
	What makes the difference	
	• Level 1 names categories without examples; Level 3 gives specific, accurate examples in every category.	
	• Level 3 includes ecosystem services, genetic resources and pest control that Level 1 omits.	
	• 'Discuss' at the top level means weighing the arguments against each other - Level 3 evaluates which arguments persuade and why; Level 1 does not.	
10.2	1 mark for the point below. **1** The variety and abundance of life (species, genes, habitats) in an area / on Earth **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Importance:** resources (timber, fuel, food, fibres, medicines) and ecosystem services (atmospheric regulation, carbon store, hydrological regulation, climate, soil conservation, habitat, recreation) • **Causes:** agriculture (subsistence and cash crops), mineral extraction, reservoirs, urbanisation, transport, unsustainable timber, fuel and fodder exploitation above MSY – trends by region • **Consequences:** loss of resources, biodiversity and species, fragmentation – hydrological changes – soil impacts – climate impacts (albedo, carbon, rainfall) • **Productivity vs biodiversity:** non-indigenous species, monocultures, close planting, simple age structure, clear-felling • **Sustainable management:** MSY harvest rates, mixed indigenous species, mixed age structure, selective logging, traditional coppicing / pollarding, FSC certification, community involvement, protected areas, debt-for-nature, afforestation programmes (China) – Rio Bravo case study	25
	• **Evaluation:** economics of slow-growing hardwoods, enforcement, demand from MEDCs, ITTO failures – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-57, 360-369, 386-387; AQA spec 3.5.3, 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Why monitor:** species present, population size, density, distribution, survival, age structure, breeding rates, movements – baseline and evaluation – Red List categorisation and quotas rely on it • **Standard techniques and their limitations:** quadrats, transects, nets, traps, Lincoln index, Simpson's index, DAFOR – sampling design (random / systematic, size, number, timing, standardisation) • **New technologies:** satellite / radio / GPS / acoustic tracking, geolocators, camera traps, CCTV, image recognition databases, DNA databases, eDNA, sonograms, drones / ROVs / AUVs, satellite sensors, data loggers • **How they improve validity:** representative data over large areas and times, less disturbance, individual identification, detection of hidden species, objectivity • **Examples:** penguin colonies from satellites, tiger camera traps, ivory DNA, MCZ monitoring, released animals tracked	25
	• **Limitations:** cost, battery / tag weight, data handling, ethics, still need ground truthing • Judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-407; AQA spec 3.1.2.3.5-3.1.2.3.6, 3.7.	

Do not write outside the box